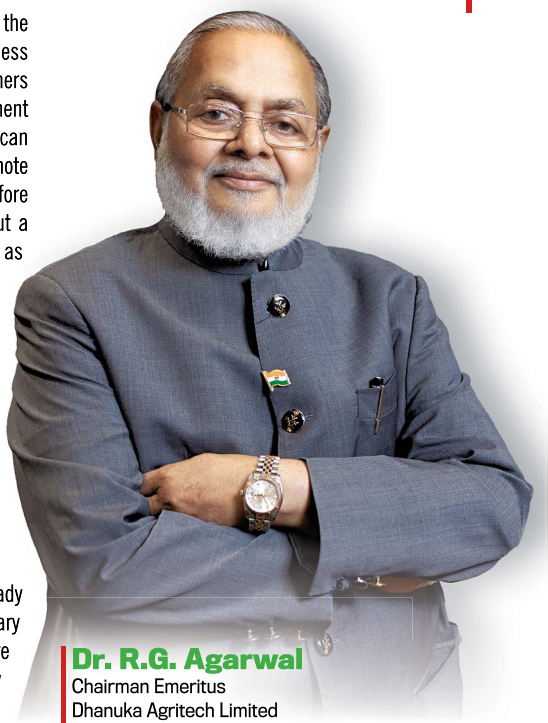


Dr. R.G. Agarwal
Chairman Emeritus
Dhanuka Agritech Limited
groupchairmanoffice@dhanuka.com

Exacerbating this challenge is a significant deficiency in awareness. Even in contexts where modern technologies are available, a significant proportion of farmers remain oblivious of their existence or are uncertain about how to implement them effectively. Government survey data reveal that approximately 60 percent of farming households did not obtain information on modern agricultural technologies from any formal source, resulting in a significant portion remaining disconnected from innovations that could enhance productivity and income levels. Reliable government and ministry estimate further indicate that approximately 40 to 45 percent of producers are acquainted with advanced agricultural technologies. This deficiency in awareness is equally detrimental as the lack of technology itself, since innovation holds limited value unless it is accessible to producers in a manner that is comprehensible, cost-effective, and reliable.

The issue is further compounded by the extensive proliferation of substandard and counterfeit agricultural inputs. Counterfeit seedlings, contaminated fertilisers, and counterfeit pesticides have emerged as a covert epidemic in many regions of the country. These inputs not only diminish yields; they also devastate crops, pose health risks to human and environment, undermine farmers' confidence in scientific and institutional authorities, and drive families into debt. Recent enforcement actions in states including Uttar Pradesh, Madhya Pradesh, Gujarat, Haryana, and Telangana have revealed extensive interstate operations involved in the production and distribution of hundreds of thousands of counterfeit fertilisers stocks each day. The seriousness of the issue is highlighted by the Uttar Pradesh government's decision to invoke the National Security Act against individuals engaged in fertiliser counterfeiting and black marketing. This constitutes not merely an economic offence but also poses a direct threat to India's food and nutritional security.

India currently maintains robust regulatory frameworks, including the Seeds Act, the Fertiliser Control Order, and the Insecticides Act, complemented by ongoing reforms such as the proposed PMB 2020. The primary difficulty lies not in the lack of legislation, but in their inadequate and inconsistent enforcement. When enforcement is weak, unlawful operators prosper, legitimate enterprises get adversely affected, and farmers ultimately incur the repercussions. What is urgently needed is rigorous enforcement complemented by digital traceability, QR-code-based authentication, accredited laboratories, and prompt, exemplary sanctions for repeat and deliberate offenders. Protecting farmers from substandard and counterfeit inputs is essential to restoring trust and improving productivity.

Another significant structural obstacle is the restricted scope of farmer education, awareness and extension services. With over 14 crore farmers distributed across 6.5 lakh villages, no government mechanism, regardless of its good intentions, can independently reach farmers in its most remote villages. Public–private partnerships are therefore essential, no longer a matter of coincidence but a deliberate choice for a progressive nation such as India. The private sector, leveraging its capacity for innovation, extensive field networks, and digital platforms, must be supported and incentivised to augment public initiatives in research, training, and the dissemination of technology. When public institutions and accountable private enterprises collaborate effectively, producers benefit from advanced scientific knowledge, dependable markets, and innovative technologies.

Encouragingly, collaborative models are already illustrating the potential outcomes. Contemporary advanced crop protection technologies are considerably safer, more precise, and markedly more efficient compared to those of earlier times. Pesticides that previously necessitated application in large volumes are now effective in few grams, providing increased efficacy with a significantly reduced environmental impact. Precision technologies such as drones, weather monitoring stations, soil and agricultural sensors, and data-driven advisory services are empowering farmers to optimise input utilisation, lower expenses, and enhance crop yields. However, these technologies remain confined to limited pockets and have yet to reach the majority of Indian farmers.

A vital driver of transformation is investment in research and development, where India persistently underperforms. National R&D expenditure stands at about 0.7 per cent of GDP, compared with nearly 2.6 per cent and 5.6 per cent in innovation-driven economies such as China and Israel, respectively. Ironically the Prime Minister's inspiring call of "Jai Anusandhan" cannot be realised without sustained and meaningful investment in agricultural education and R&D. Earlier incentives, such as the 200 percent rebate in income tax for R&D expenditure by DST-recognized companies, significantly contributed to promoting private sector innovation. Their withdrawal has diminished research funding during a period of heightened global competition and warrants urgent revisit.

India's national ambitions are ambitious and compelling. The goal of establishing a USD 5 trillion economy by 2028, with agriculture accounting for nearly USD 1 trillion, and a USD 30 trillion Viksit Bharat by 2047, cannot be realised without a fundamental transformation of the agricultural sector. India's developmental path will inevitably traverse its agricultural territories, farmlands, and rural settlements. Agriculture must transition from reliance on subsidies to a sector characterised by productivity, resilience, and entrepreneurship, where producers increase their earnings per acre through access to advanced technologies, reliable quality inputs, efficient markets, and value addition.

The path ahead involves transitioning from short-term interventions to sustainable, long-term capacity development. Farmers should be safeguarded against fraud, equipped with knowledge, and provided with modern instruments. Awareness, education, and trust are pivotal as the technology itself. When producers receive proper training, information, and develop confidence, they transition into entrepreneurs rather than remaining perpetual recipients of subsidies. In doing so, agriculture can develop into a true driver of national development.

India possesses the land, the soil, the scientific expertise, the industry, and most importantly, the character of its farmers. What is currently needed is a system that assures the accessibility of innovative and advanced technologies, guarantees the quality of agricultural inputs, and addresses the awareness gap at the final stage. If India succeeds in addressing these constraints, agriculture will no longer be a story of unrealised promise. It will not only nourish the nation but also fuel India's ascent as a leading global economic power.

